

Electronic Locking Differential (ELD) System

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST2574-A	Flex Probe Kit 300-NUD105-R025DE or equivalent
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

Electronic Locking Differential (ELD) System

The ELD system allows the operator to lock the rear differential side gears. When the ELD is engaged both rear wheels turn at the same speed, providing added traction on slippery and/or off road surfaces. The operator can engage the ELD per the charts below:

ELD Engagement Speeds — 4WD Vehicles

Switch Position	Engage	Disengage	Re-engage
2H/4A/4H	31 km/h (19 mph) ^a	42 km/h (25 mph)	31 km/h (19 mph) ^a
4L	90 km/h (56 mph) ^a	100 km/h (62 mph)	90 km/h (56 mph) ^a

^aThe ELD system engages/re-engages at or below the speed noted.

ELD Engagement Speeds — 2WD Vehicles

Switch Position	Engage	Disengage	Re-engage
Off	N/A	N/A	N/A
On	31 km/h (19 mph) ^a	42 km/h (25 mph)	31 km/h (19 mph) ^a

^aThe **ELD** system engages/re-engages at or below the speed noted.

When the operator activates the **ELD** , the **ELD** icon displays in the **IPC** message center. If the vehicle is driven above the disengage speed, the **IPC** message center no longer displays the **ELD** icon, alerting the operator that the **ELD** is disengaged. Once the vehicle drops below the re-engage speed, the **IPC** message center displays the **ELD** icon again. If there is a malfunction with the **ELD** system, a "CHECK LOCKING DIFFERENTIAL" message is displayed in the **IPC** message center, letting the operator know the **ELD** is not engaged.

The differential locks when a **PWM** controlled magnetic field coil located inside the differential energizes. A collar within the differential moves toward a differential side gear and engages a dog clutch within the differential. This locks the differential side gears and creates the locked differential.

The **TCCM** activates the **ELD** . When the **ELD** is engaged, the **TCCM** sends **PWM** voltage to the **ELD** field coil located in the differential. This voltage is initially 98% duty cycle, and then tapers off to 40% duty cycle after 30 seconds. In order for the **TCCM** to activate the **ELD** , certain conditions must be met.

TCCM Input Conditions

- **MSS** set to **ELD** position
- Average rear wheel speed versus average front wheel speed must be less than 7 km/h (4.3 mph) (via **HS-CAN** communication from ABS module)
- Operating range (voltage between 9-16 volts)
- DTC setting range (voltage between 10-15 volts)
- Ignition in the ON position
- Throttle Position (TP) less than 50%

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

• Rear axle assembly • Matching tire sizes	• Battery Junction Box (BJB) fuses: ▪ 20 (20A) ▪ 29 (10A) ▪ 54 (5A) ▪ 68 (25A) • Body Control Module (BCM) fuse 34 (10A) (vehicles with ELD only) • Wiring, terminals or connectors • Mode Select Switch (MSS) • Electronic Locking Differential (ELD) field coil • Transfer Case Control Module (TCCM)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the **DLC** .

5. **NOTE:** *The **BCM** LED prove out confirms power and ground from the **DLC** are provided to the **BCM** .*

If the scan tool does not communicate with the **BCM** :

- check the **BCM** connection to the vehicle.
- check the scan tool connection to the **BCM** .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:

- verify the ignition is in the ON position.
- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00 to diagnose no response from the PCM.

7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
- If the network test passes, retrieve and record **CM**DTCs .

8. Carry out the self-test diagnostics for the **TCCM** .

9. If the DTCs retrieved are related to the concern, go to the **TCCM** DTC Chart in this section. For all other DTCs, refer to Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart — Electronic Locking Differential \(ELD\) System](#) in this section.

DTC Charts

Transfer Case Control Module (TCCM) DTC Chart — **ELD** System

NOTE: *This module utilizes a 5-character DTC followed by a 2-character failure type code. The failure type code provides information about specific fault conditions such as opens or shorts to ground. have an additional 2-character DTC status code suffix to assist in determining DTC history.*

NOTE: *Network DTCs (U-codes) are often a result of intermittent concerns such as damaged wiring or low battery voltage occurrences. Additionally, vehicle repair procedures such as module reprogramming often set network DTCs. Replacing a module to resolve a network DTC is unlikely to resolve the concern. To prevent repeat network DTC concerns, inspect all network wiring, especially connectors. Test the vehicle battery, refer to Section 414-01.*

DTC	Description	Action
P0562:00	System Voltage Low: No Sub Type Information	GO to Pinpoint Test M.
P0563:00	System Voltage High: No Sub Type Information	GO to Pinpoint Test M.
P185A:00	Differential Lock-Up Input Switch Circuit High: No Sub Type Information	GO to Pinpoint Test Y.
P185B:00	Differential Lock-Up Indicator Circuit Low: No Sub Type Information	GO to Pinpoint Test Z.
P185C:00	Differential Lock-Up Indicator Circuit High: No Sub Type Information	GO to Pinpoint Test AA.
P185D:00	Differential Lock-Up Control Circuit Performance: No Sub Type Information	GO to Pinpoint Test AB.
P187C:00	Differential Lock-Up Control Circuit/Open: No Sub Type Information	GO to Pinpoint Test AC.
P187D:00	Differential Lock-Up Control Circuit Low: No Sub Type Information	GO to Pinpoint Test AC.
P187E:00	Differential Lock-Up Control Circuit High: No Sub Type Information	GO to Pinpoint Test AD.
U0100:00	Lost Communication With ECM/PCM "A": No Sub Type Information	GO to Pinpoint Test AE.
U0121:00	Lost Communication With Anti-Lock Brake System (ABS) Control Module: No Sub Type Information	GO to Pinpoint Test AF.
U0140:00	Lost Communication With Body Control Module (BCM): No Sub Type Information	GO to Pinpoint Test AG.
U0418:00	Invalid Data Received From Brake System Control Module: No Sub Type Information	REFER to Section 206-09 to diagnose the wheel speed sensor concern.
U2300:54	Central Configuration: Missing Calibration	This DTC can set if one or more expected configuration data identifications is missing (gearbox type, tire circumference) from the BCM . CARRY OUT a self-test of the BCM . If DTCs are present, REFER to Section 419-10. If no DTCs are present, CLEAR the DTCs. CARRY OUT a self-test. If U2300:54 returns, INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section.
U2300:55	Central Configuration: Not Configured	This DTC can set upon power up only until all expected configuration data parameters (gearbox type, axle ratio, tire circumference) is received from the BCM . CARRY OUT a self-test of the BCM . If DTCs are present, REFER to

DTC	Description	Action
		Section 419-10. If no DTCs are present, CLEAR the DTCs. CARRY OUT self-test. If U2300:55 returns, INSTALL a new <u>TCCM</u> . REFER to Transfer Case Control Module (TCCM) in this section.
U2300:56	Central Configuration: Invalid / Incompatible Configuration	This DTC can set if one expected configuration data parameter is out of range (gearbox type, axle ratio, tire circumference) from the <u>BCM</u> . CARRY OUT a self-test of the <u>BCM</u> . If DTCs are present, REFER to Section 419-10. If no DTCs are present, CLEAR the DTCs. CARRY OUT a self-test. If U2300:56 returns, INSTALL a new <u>TCCM</u> . REFER to Transfer Case Control Module (TCCM) in this section.
U2300:64	Central Configuration: Signal Plausibility Failure	This DTC can set if the expected axle ratio or tire circumference is out of range from the <u>BCM</u> . CARRY OUT a self-test of the <u>BCM</u> . If DTCs are present, REFER to Section 419-10. If no DTCs are present, CLEAR the DTCs. CARRY OUT a self-test. If U2300:64 returns, INSTALL a new <u>TCCM</u> . REFER to Transfer Case Control Module (TCCM) in this section.
U3000:41	Control Module: General Checksum Failure	CLEAR the DTC. REPEAT the self-test. If DTC U3000:41 is retrieved, INSTALL a new <u>TCCM</u> . REFER to Transfer Case Control Module (TCCM) in this section.
U3000:46	Control Module: Calibration / Parameter Memory Failure	CLEAR the DTC. REPEAT the self-test. If DTC U3000:46 is retrieved, INSTALL a new <u>TCCM</u> . REFER to Transfer Case Control Module (TCCM) in this section.
U3000:49	Control Module: Internal Electronic Failure	CLEAR the DTC. REPEAT the self-test. If DTC U3000:49 is retrieved, INSTALL a new <u>TCCM</u> . REFER to Transfer Case Control Module (TCCM) in this section.
U300D:11	Ignition Input On/Start: Circuit Short To Ground	GO to Pinpoint Test L.
All other <u>TCCM</u> DTCs	—	REFER to the <u>TCCM</u> DTC Chart in of this section.

Symptom Chart — Electronic Locking Differential (ELD) System

Symptom Chart		
Condition	Possible Causes	Action
No communication with the <u>TCCM</u>	- Wiring, terminals or connectors - Scan tool - DLC <u>TCCM</u>	CHECK the module communications network. REFER to Section 418-00.
Rear wheels binding when turning corners with the <u>MSS</u> in 2WD or OFF	- Wiring, terminals or connectors <u>ELD</u> field coil <u>TCCM</u> - Rear differential	GO to Pinpoint Test AD.

Pinpoint Tests

Pinpoint Test Y: DTC P185A:00

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

When the MSS is engaged to lock the rear differential, the TCCM sends a voltage signal through the MSS to ground pulling the voltage low. When the TCCM senses the voltage is low it activates the ELD .

- DTC P185A:00 (Differential Lock-Up Input Switch Circuit High: No Sub Type Information) — When not in locking mode, the TCCM senses whether the ELD mode switch circuit has low voltage and if the differential is always engaged. This DTC will not set unless the MSS is set to 2H or OFF. If the ELD is functioning normally, the differential will immediately lock when the MSS is activated and the ELD operating conditions are qualified. This DTC will only display during the self-test function.

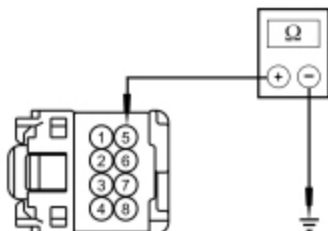
This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- MSS
- TCCM

PINPOINT TEST Y : DTC P185A:00

NOTE: Make sure the battery is fully charged before carrying out this pinpoint test.	
Y1 CHECK FOR TCCM DTCS	
NOTE: Failure to place the MSS in 2H or OFF prior to carrying out a self-test of the TCCM (to disable the ELD field coil) results in an inaccurate self-test.	
• Ignition ON. • Set the MSS in 2H (ESQF or 2 speed-torque-on-demand). • Set the MSS in OFF (2WD). • Enter the following diagnostic mode on the scan tool: Self-Test — TCCM .	
Are any DTCs present?	
Yes	If only DTC P185A:00 is retrieved, GO to Y2 . For any other DTCs, REFER to the TCCM DTC Chart in this section.
No	IDENTIFY the customer concern and GO to Symptom Chart — Electronic Locking Differential (ELD) System in this section.

Y2 CHECK FOR A SHORT TO GROUND ON THE LOCKER MODE SWITCH CIRCUIT	
• Ignition OFF. • Disconnect: TCCM C2371A . • Disconnect: MSS C284 . • Measure the resistance between MSS C284 Pin 5, , circuit CCF32 (WH/GN), harness side and ground.	



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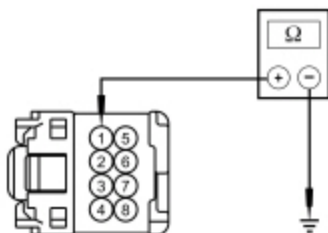
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Is the resistance greater than 10,000 ohms?

Yes	GO to Y3 .
No	REPAIR the circuit..

Y3 CHECK THE LOCKER MODE SWITCH GROUND CIRCUIT

- Ignition OFF.
- Measure the resistance between MSS [C284](#) Pin 1, circuit GD133 (BK), harness side and ground.



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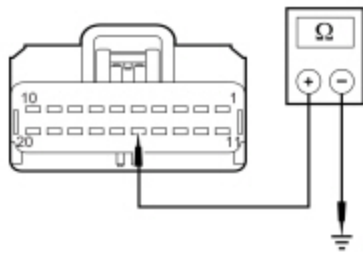
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Is the resistance less than 3 ohms?

Yes	GO to Y4 .
No	REPAIR the circuit.

Y4 CHECK FOR A STUCK MSS

- Connect: MSS [C284](#) .
- Activating and deactivating the MSS , measure the resistance between TCCM [C2371A](#) Pin 15, circuit CCF32 (WH/GN), harness side and ground.



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Is the resistance less than 3 ohms with the **MSS** activated and greater than 10,000 ohms with the **MSS** deactivated?

Yes	GO to Y5
No	INSTALL a new MSS . REFER to Mode Select Switch (MSS) in this section. TEST the system for normal operation.

Y5 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all **TCCM** connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all **TCCM** connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.

Pinpoint Test Z: DTC P185B:00

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

When the **MSS** is engaged to lock the rear differential, a **PWM** signal is sent to the **ELD** field coil by the **TCCM** through the differential lock control circuit. This **PWM** signal is initially 98% duty cycle to lock the differential and decreases to 40% duty cycle once the differential is locked. If the differential does not lock, the **TCCM** continues to send the signal for a set amount of time. Once the differential is locked, the indicator will illuminate steady.

- DTC P185B:00 (Differential Lock-Up Indicator Circuit Low: No Sub Type Information) — When not in locking mode, the TCCM C2371A-20 senses the indicator circuit is shorted to ground and the indicator is always illuminated. This DTC should not set for an open circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- TCCM

PINPOINT TEST Z : DTC P185B:00

NOTE: Make sure the battery is fully charged before performing this pinpoint test.

Z1 CHECK THE TCCM 4WD SWITCH STATUS (4X4_SW) PID

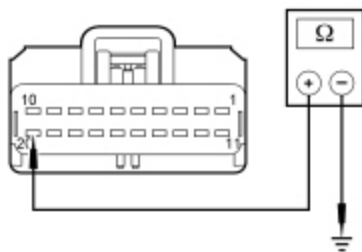
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Set the MSS to 4H and pull the knob to lock the rear differential.
- Monitor the 4X4_SW PID.

Does the 4X4_SW PID indicate 4H mode?

Yes	GO to Z2 .
No	INSTALL a new MSS . REFER to Mode Select Switch (MSS) in this section. CLEAR the DTCs. TEST the system for normal operation.

Z2 CHECK THE REAR ELD INDICATOR CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: TCCM [C2371A](#) .
- Disconnect: MSS .
- Measure the resistance between TCCM [C2371A](#) Pin 20, circuit CCF33 (GY/YE), harness side and ground.



• N0090963

Is the resistance greater than 10,000 ohms?

Yes	GO to Z3 .
No	REPAIR circuit CCF33 (GY/YE). CLEAR the DTCs. TEST the system for normal operation.

Z3 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all **TCCM** connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all **TCCM** connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.

Pinpoint Test AA: DTC P185C:00

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

When the **MSS** is engaged to lock the rear differential, a **PWM** signal is sent to the **ELD** field coil by the **TCCM** through the differential lock control circuit. This **PWM** signal is initially 98% duty cycle to lock the differential and decreases to 40% duty cycle once the differential is locked. When the differential is fully locked, the indicator will illuminate steady.

- DTC P185C:00 (Differential Lock-Up Indicator Circuit High: No Sub Type Information) — When the driver engages the **ELD** the **TCCM** C2371A-20 senses the indicator circuit has high voltage and is not illuminated.

This pinpoint test is intended to diagnose the following:

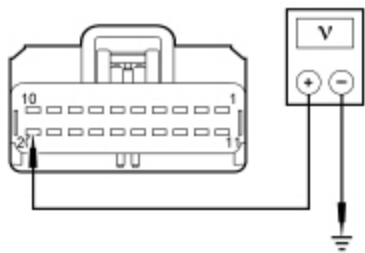
- Wiring, terminals or connectors
- **TCCM**

PINPOINT TEST AA : DTC P185C:00

NOTE: Make sure the battery is fully charged before performing this pinpoint test.	
AA1 OPERATE THE ELD INDICATOR USING THE TCCM REAR LOCKING DIFF LAMP OUTPUT CONTROL OPTIONAL (DIFF_LAMP_OS) PID	
• Enter the following diagnostic mode on the scan tool: DataLogger — TCCM . • Command the DIFF_LAMP_OS PID on.	
Does the ELD indicator in the MSS illuminate?	
Yes	GO to AA5 .
No	GO to AA2 .

AA2 CHECK THE ELD INDICATOR CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ~~TCCM~~ [C2371A](#) .
- Disconnect: ~~MSS~~ .
- Ignition ON.
- Measure the voltage between ~~TCCM~~ [C2371A](#) Pin 20, circuit CCF33 (GY/YE), harness side and ground.



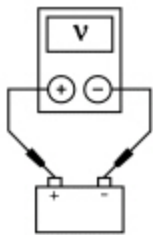
• N0090961

Is any voltage present?

Yes	REPAIR circuit CCF33 (GY/YE). CLEAR the DTCs. TEST the system for normal operation.
No	GO to AA3 .

AA3 CHECK FOR VOLTAGE TO THE MSS

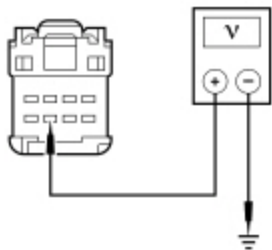
- Measure and record the battery voltage at the battery.



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- Measure the voltage between ~~MSS~~ [C284](#) Pin 7, circuit CBP34 (VT/BN), harness side and ground.



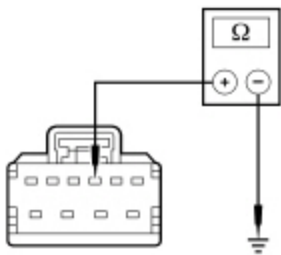
• N0090956

Does the voltage equal the recorded battery voltage?

Yes	GO to AA4 .
No	VERIFY BCM fuse 34 (10A) is OK. If OK, REPAIR circuit CBP34 (VT/BN). If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. TEST the system for normal operation.

AA4 CHECK THE TCCM GROUND CIRCUIT

- Ignition OFF.
- Disconnect: TCCM [C2371B](#) .
- Measure the resistance between [C2371B](#) Pin 3, circuit GD123 (BK/GY), harness side and ground.



• N0090962

Is the resistance less than 3 ohms?

Yes	GO to AA5 .
No	REPAIR circuit GD123 (BK/GY). CLEAR the DTCs. TEST the system for normal operation.

AA5 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all TCCM connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all TCCM connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.

Pinpoint Test AB: DTC P185D:00

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

When the **MSS** is engaged to lock the rear differential, a **PWM** signal is sent to the **ELD** field coil by the **TCCM** through the differential lock control circuit. This **PWM** signal is initially 98% duty cycle to lock the differential and decreases to 40% duty cycle once the differential is locked. If the differential does not lock, the **TCCM** continues to send the signal for a set amount of time. Once the differential is locked, the indicator will illuminate steady.

- DTC P185D:00 (Differential Lock-up Control Circuit Performance: No Sub Type Information) — The **TCCM** sends a signal to the lock up solenoid to lock the differential. After several unsuccessful attempts, the **TCCM** sets this DTC alerting the driver the **ELD** is inoperative. Fault may be due to a wheel speed condition or due to a mechanical failure within the driveline/axle subsystem.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- **ELD** field coil

PINPOINT TEST AB : DTC P185D:00

NOTE: Make sure the battery is fully charged before carrying out this pinpoint test.

AB1 CHECK THE ABS MODULE WHEEL SPEED SENSOR PIDS

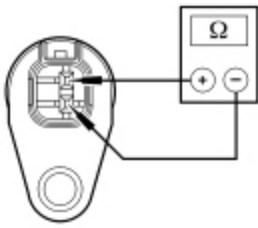
- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .
- While driving the vehicle at 30 km/h (18 mph), monitor the following wheel speed sensor PIDs:
 - Left Front Wheel Speed Sensor (LF_WSPD)
 - Left Rear Wheel Speed Sensor (LR_WSPD)
 - Right Front Wheel Speed Sensor (RF_WSPD)
 - Right Rear Wheel Speed Sensor (RR_WSPD)
- Compare the speedometer reading to the wheel speed sensor PIDs.

Do the wheel speed sensor PIDs and speedometer speeds match within 1.5 km/h (1 mph)?

Yes	GO to AB2 .
No	VERIFY that the 4 road wheels are the same size and have similar tread depth. CHECK the ABS for faults. REFER to Section 206-09. TEST the system for normal operation.

AB2 CHECK RESISTANCE OF THE ELD FIELD COIL

- Ignition OFF.
 - Disconnect: **ELD** Field Coil .
 - **NOTE:** If the temperatures are exceptionally high, a higher than normal resistance reading could result. A lower temperature could result in a lower resistance reading.
- Measure the resistance between **ELD** field coil C4359 pins 1 and 2, component side.



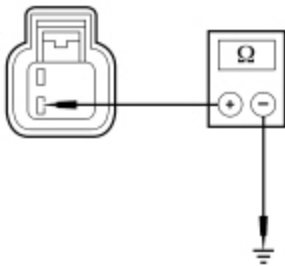
• N0091387

Is the resistance between 2-4 ohms?

Yes	GO to AB3 .
No	INSTALL a new <u>ELD</u> field coil. REFER to Section 205-02B. CLEAR the DTCs. TEST the system for normal operation.

AB3 CHECK THE ELD FIELD COIL GROUND

- Measure the resistance between ELD field coil [C4359](#) Pin 2, circuit GD123 (BK/GY), harness side and ground.



• N0090955

Is the resistance less than 3 ohms?

Yes	REFER to Section 205-02B to repair the rear differential. CLEAR the DTCs. TEST the system for normal operation.
No	REPAIR circuit GD123 (BK/GY). CLEAR the DTCs. TEST the system for normal operation.

Pinpoint Test AC: DTC P187C:00 and P187D:00

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

When the MSS is engaged to lock the rear differential, a PWM signal is sent to the ELD field coil by the TCCM through the differential lock control circuit. This PWM signal is initially 98% duty cycle to lock the differential and decreases to 40% duty cycle once the differential is locked. If the differential lock control circuit is open or shorted to ground, the TCCM senses a signal is not reaching the differential and sets a DTC.

- DTC P187C:00 (Differential Lock-Up Control Circuit/Open: No Sub Type Information) — The TCCM sends a signal to the differential to lock the differential. The TCCM C2371B pin 9 senses that the differential lock control circuit is

open. The **ELD** differential is not engaged.

- DTC P187D:00 (Differential Lock-Up Control Circuit Low: No Sub Type Information) — The **TCCM** sends a signal to lock the differential. The **TCCM** C2371B pin 9 senses that the differential lock control circuit is shorted to ground. The **ELD** differential is not engaged.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- **TCCM**

PINPOINT TEST AC : DTC P187C:00 AND P187D:00

NOTE: Make sure the battery is fully charged before carrying out this pinpoint test.

AC1 CHECK FOR TCCM DTCS

NOTE: Failure to place the **MSS** in 2H or OFF prior to carrying out a self-test of the **TCCM** (to disable the **ELD** field coil) results in inaccurate self-test.

- Ignition ON.
- Set the **MSS** in 2H (**ESOF** or 2 speed-torque-on-demand).
- Set the **MSS** in OFF (**2WD**).
- Enter the following diagnostic mode on the scan tool: Self-Test — **TCCM** .

Are any DTCs present?

Yes	For DTC P187C:00, GO to AC4 . For DTC P187D:00, GO to AC2 . For all other DTCs, REFER to the TCCM DTC Chart in this section.
No	IDENTIFY the customer concern and GO to Symptom Chart — Electronic Locking Differential (ELD) System in this section.

AC2 CHECK FOR TCCM DTCS WITH THE ELD FIELD COIL DISCONNECTED

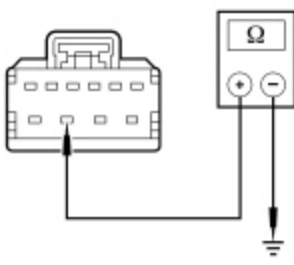
- Ignition OFF.
- Disconnect: **ELD** Field Coil .
- Ignition ON.
- Set the **MSS** in 2H (**ESOF** or 2 speed-torque-on-demand).
- Set the **MSS** in OFF (**2WD**).
- Enter the following diagnostic mode on the scan tool: Self-Test — **TCCM** .

CLEAR the DTCs. REPEAT the self-test. Does DTC P187D:00 return?

Yes	GO to AC3 .
No	INSTALL a new field coil. REFER to Section 205-02B. CLEAR the DTCs. TEST the system for normal operation.

AC3 CHECK THE ELD DIFFERENTIAL LOCK LOCK CONTROL CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: ~~TCCM~~ [C2371B](#) .
- Measure the resistance between ~~TCCM~~ [C2371B](#) Pin 9, circuit CCF31 (BN/VT), harness side and ground.



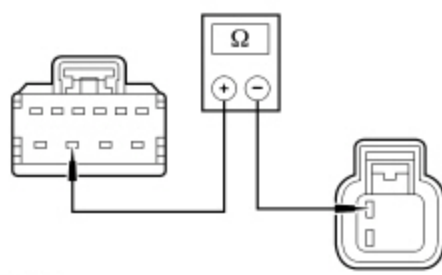
• N0090960

Is the resistance greater than 10,000 ohms?

Yes	GO to AC4 .
No	REPAIR circuit CCF31 (BN/VT). CLEAR the DTCs. TEST the system for normal operation.

AC4 CHECK THE ELD DIFFERENTIAL LOCK CONTROL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: ~~TCCM~~ [C2371B](#) .
- Disconnect: ~~ELD~~ Field Coil .
- Measure the resistance between ~~TCCM~~ [C2371B](#) Pin 9, circuit CCF31 (BN/VT), harness side and ~~ELD~~ field coil [C4359](#) Pin 1, circuit CCF31 (BN/VT), harness side.



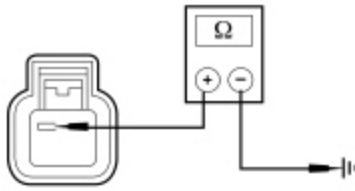
• N0090956

Is the resistance less than 3 ohms?

Yes	GO to AC5 .
No	REPAIR circuit CCF31 (BN/VT). CLEAR the DTCs. TEST the system for normal operation.

AC5 CHECK THE ELD DIFFERENTIAL LOCK GROUND CIRCUIT FOR AN OPEN

- Measure the resistance between ~~ELD~~ [C4359](#) Pin 2, harness side and ground



- N0165201

Is the resistance less than 3 ohms?

Yes	INSTALL a new <u>ELD</u> field coil. If concern still exists, GO to AC6 .
No	REPAIR the circuit. CLEAR the DTCs.

AC6 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all TCCM connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all TCCM connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new <u>TCCM</u> . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.

Pinpoint Test AD: Rear Wheels Binding When Turning Corners With the ELD OFF

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

When the MSS is engaged to lock the rear differential, a PWM signal is sent to the ELD field coil by the TCCM through the differential lock control circuit. This PWM signal is initially 98% duty cycle to lock the differential and decreases to 40% duty cycle once the differential is locked. If the differential does not lock, the TCCM continues to send the signal for a set amount of time. Once the differential is locked, the indicator will illuminate steady.

- DTC P187E:00 (Differential Lock-Up Control Circuit High: No Sub Type Information) — The TCCM sends a signal to lock the differential. This DTC sets when the TCCM C2371B-9 senses that the differential lock control circuit is

shorted to battery voltage. The differential is constantly engaged.

This pinpoint test is intended to diagnose the following:

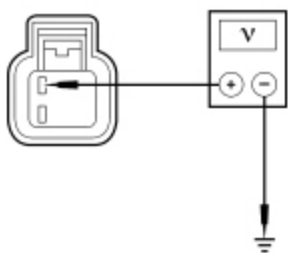
- Wiring, terminals or connectors
- ~~ELD~~ field coil
- ~~TCCM~~

PINPOINT TEST AD : REAR WHEELS BINDING WHEN TURNING CORNERS WITH THE ELD OFF

NOTE: Make sure the battery is fully charged before carrying out this pinpoint test.

AD1 CHECK FOR VOLTAGE TO THE ELD FIELD COIL

- Ignition OFF.
- Disconnect: ~~ELD~~ Field Coil .
- Ignition ON.
- Set the MSS in 2H (~~ESOF~~ or 2 speed-torque-on-demand).
- Set the MSS in OFF (~~2WD~~).
- Measure the voltage between ~~ELD~~ field coil [C4359](#) Pin 1, circuit CCF31 (BN/VT), harness side and ground.



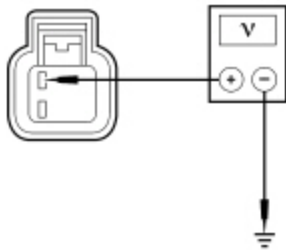
• N0090954

Is any voltage present?

Yes	GO to AD2 .
No	REFER to Section 205-02B to repair the rear differential. CLEAR the DTCs. TEST the system for normal operation.

AD2 CHECK THE ELD FIELD COIL CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ~~TCCM~~ [C2371B](#) .
- Ignition ON.
- Measure the voltage between field coil [C4359](#) Pin 1, circuit CCF31 (BN/VT), harness side and ground.



• N0090954

Is any voltage present?

Yes	REPAIR circuit CCF31 (BN/VT). CLEAR the DTCs. TEST the system for normal operation.
No	GO to AD3 .

AD3 CHECK THE TCCM REAR LOCKING DIFFERENTIAL PWM OUTPUT CONTROL (DIFF_PWM_OS) PID

- Ignition OFF.
- Connect: ELD Field Coil .
- Connect: TCCM [C2371B](#) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- With the MSS in 2H or OFF (depending upon vehicle option content), monitor the DIFF_PWM_OS PID.

Does the DIFF_PWM_OS PID read OFF?

Yes	GO to AD4 .
No	INSTALL a new <u>MSS</u> . REFER to Mode Select Switch (MSS) in this section. CLEAR the DTCs. TEST the system for normal operation.

AD4 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all TCCM connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all TCCM connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.

Pinpoint Test AE: DTC U0100:00

Normal Operation

- DTC U0100:00 (Lost Communication With ECM/PCM "A": No Sub-Type Information) — sets in continuous memory in the **TCCM** if data messages received from the PCM over the **HS-CAN** are missing for 5 seconds or longer with ignition in RUN and battery voltage in range. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- PCM
- **TCCM**

PINPOINT TEST AE : DTC U0100:00

AE1 CHECK THE TCCM CONTINUOUS MEMORY DTCS	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: TCCM Self-Test . • Clear the DTCs. • Wait 10 seconds. • Repeat the TCCM self-test.	
Is DTC U0100:00 retrieved again?	
Yes	GO to AE2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
AE2 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST	
• Check for recorded DTCs from the PCM KOEO self-test.	
Is DTC P0562:00 or P0563:00 recorded?	
Yes	REFER to Section 414-00.
No	GO to AE3 .
AE3 RETRIEVE THE RECORDED DTCS FROM THE TCCM SELF-TEST	
• Check for recorded DTCs from the TCCM self-test.	

Is DTC P0562:00 or P0563:00 recorded?

Yes	REFER to GO to Pinpoint Test M.
No	GO to AE4 .

AE4 CHECK FOR DTC U0100:00 SET IN OTHER MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — All CMDTCs .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0100:00 in the ~~RCM~~ , or DTC U0100:87 in the ~~BCM~~ ?

Yes	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.
No	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.

Pinpoint Test AF: DTC U0121:00

Normal Operation

- DTC U0121:00 (Lost Communication With Anti-Lock Brake System (ABS) Control Module: No Sub-Type Information) — sets in continuous memory in the ~~TCCM~~ if data messages received from the ABS module over the ~~HS-CAN~~ are missing for 5 seconds or longer with ignition in RUN and battery voltage in range. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- ABS module
- ~~TCCM~~

PINPOINT TEST AF : DTC U0121:00

AF1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AF2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AF2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the ABS module pass the network test?

Yes	GO to AF3 .
No	REFER to Section 418-00.

AF3 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE SELF-TEST

- Check for recorded DTCs from the ABS module self-test.

Is DTC B11E8:16 (B1318) or DTC B11E8:17 (B1317) recorded?

Yes	REFER to Section 206-09.
No	GO to AF4 .

AF4 RETRIEVE THE RECORDED DTCS FROM THE TCCM SELF-TEST

- Check for recorded DTCs from the TCCM self-test.

Is DTC P0562:00 or P0563:00 recorded?

Yes	GO to Pinpoint Test M.
No	GO to AF5 .

AF5 RECHECK THE TCCM DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the PMI or the PMI may not have been carried out.*

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the TCCM self-test.

Is DTC U0121:00 still present?

Yes	INSTALL a new ABS module. REFER to Section 206-09. CLEAR the DTCs. REPEAT the <u>TCCM</u> self-test. If DTC U0121:00 is still present, INSTALL a new <u>TCCM</u> . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

Pinpoint Test AG: DTC U0140:00

Normal Operation

- DTC U0140:00 (Lost Communication With Body Control Module: No Sub-Type Information) — sets in continuous memory in the TCCM if data messages received from the BCM over the HS-CAN are missing for 5 seconds or longer with ignition in RUN and battery voltage in range. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- BCM
- TCCM

PINPOINT TEST AG : DTC U0140:00

AG1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AG2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AG2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the pass the network test?

Yes	GO to AG3 .
No	REFER to Section 418-00.

AG3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded DTCs from the BCM self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	Section 419-10.
No	GO to AG4 .

AG4 RETRIEVE THE RECORDED DTCS FROM THE TCCM SELF-TEST

- Check for recorded DTCs from the TCCM self-test.

Is DTC P0562:00 or P0563:00 recorded?

Yes	GO to Pinpoint Test M.
No	GO to AG5 .

AG5 RECHECK THE TCCM DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the or the may not have been carried out.*

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the TCCM self-test.

Is DTC U0140:00 still present?

Yes	GO to AG6 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AG6 CHECK FOR DTC U0140 OR DTC U0140:00 SET IN OTHER MODULES

- Clear all DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test — All CMDTCs .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0140 or U0140:00 set in more than one module?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.

Pinpoint Test AH: Neutral Flat Tow Functional Test

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive System for schematic and connector information.

Normal Operation

Neutral flat tow operates by shifting the transfer case shift motor into the NEUTRAL position, locking the transfer case in NEUTRAL. The TCCM sends the transfer case shift motor NEUTRAL position status to the IPC over the HS-CAN .

This pinpoint test is intended to diagnose the following:

- IPC
- TCCM
- TR sensor from PCM
- Ignition switch

PINPOINT TEST AH : NEUTRAL FLAT TOW FUNCTIONAL TEST

WARNING: Shifting the transfer case to its neutral position for neutral flat towing may allow the vehicle to be able to roll even if the transmission is in park (P). Make sure the foot brake is depressed and the vehicle is in a secure and safe position while the transfer case is being shifted to its neutral position. Failure to follow this instruction may result in serious personal injury.

WARNING: Do not disconnect the battery during neutral flat towing. Doing so will prevent the transfer case from shifting properly and may allow the vehicle to roll even if the transmission is in park (P). Failure to follow this instruction may result in serious personal injury.

NOTICE: 2WD vehicles cannot be towed with any wheels on the ground as vehicle or transmission damage may occur.

NOTICE: Failing to put the transfer case in its NEUTRAL position will damage vehicle components.

NOTE: 4WD vehicles can only be flat towed with all wheels on the ground by placing the transfer case in its NEUTRAL position.

AH1 CHECK NEUTRAL FLAT TOW ACTIVATION AND DEACTIVATION

- Activate neutral flat tow. Refer to Neutral Flat Tow Activation and Deactivation in this section.

Does the IPC indicate neutral flat tow is activated?

Yes	RETURN the vehicle to the customer. ADVISE the customer about the correct Neutral Flat Tow Activation and Deactivation procedure.
No	GO to AH2 .

AH2 CHECK NEUTRAL FLAT TOW ACTIVATION WHILE MONITORING THE TCCM 4WD MODES (4WD_MODES) PID

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Monitor the 4WD_MODES PID.

Does the PID indicate NEUTRAL?

Yes	REFER to Section 413-01 to diagnose IPC faults. After the repair is complete, CLEAR the DTCs. TEST the system for normal operation.
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No	GO to AH3 .
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AH3 CHECK FOR TCCM DTCS

- Enter the following diagnostic mode on the scan tool: Self-Test — **TCCM** .

Are any DTCs present?

Yes	REFER to the TCCM DTC Chart in this section. After the repair is complete, CLEAR the DTCs. TEST the system for normal operation.
No	GO to AH4 .

AH4 CHECK TCCM BRAKE ON/OFF (4X4_BOO) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — **TCCM** .
- Monitor the 4X4_BOO PID while pressing the brake pedal.

Does the PID agree with the brake pedal position?

Yes	GO to AH5 .
No	REFER to Section 417-01 to diagnose stoplamp system faults. After the repair is complete, CLEAR the DTCs. TEST the system for normal operation.

AH5 CHECK THE TCCM IGNITION POSITION (KEY_POS_4X4) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — **TCCM** .
- Monitor the KEY_POS_4X4 PID with the ignition in the ON position.

Does the PID indicate Ignition On?

Yes	GO to AH6 .
No	REFER to Section 211-05 for further diagnosis of the ignition switch.

AH6 CHECK THE TCCM NEUTRAL INDICATOR (NEUTRAL) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — **TCCM** .
- Monitor the NEUTRAL PID with the gear selector in neutral position.

Does the PID indicate TRUE?

Yes	For vehicles equipped with the ESOF system, GO to Pinpoint Test B to diagnose the transfer case shift motor. For vehicles equipped with the 2-speed torque-on-demand system, GO to Pinpoint Test P to diagnose the transfer case shift motor.
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No	REFER to Section 307-01 for diagnosis of the TR sensor.
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